USE OF PHOTOGRAMMETRY IN MARINE ARCHAEOLOGY

PROJECT REPORT

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Abstract

Photogrammetry in archaeology has been used extensively however, it is only in recent times that photogrammetry has emerged as an important tool in marine archaeology. The underwater environment poses several challenges in photogrammetry, from the data acquisition phase to the post-processing. In this project we will explore affordable methods and tools that can be used in underwater photogrammetry for marine archaeology. We will primarily focus on the use of videos as a dataset and how adding EXIF data to the frame captures of the video can improve the 3D model.

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1. Introduction

Photogrammetry is fast becoming a key technique in archaeology for studying artefacts. Photogrammetry provides a non-invasive, low-cost and affordable method to document and measure archaeological sites and artefacts for archaeological studies. This technique can be used for recording subjects where high details are required. Photogrammetry provides geometric details of an artefact along with textural information. Photogrammetry can be defined as the science and technology of obtaining reliable information about physical objects and the environment through the process of recording, measuring and interpreting photographic images and patterns of electromagnetic radiant imagery and other phenomena.

In the last decade several studies have been done on image-based underwater photogrammetry for archaeology. However, prior studies have not shown the use of video files as the data set for photogrammetry and the importance of EXIF data like focal length and type of camera lens. In the underwater environment it can be quite challenging to shoot several quality images. Thus using videos as a data set can prove to be more convenient.
This paper has three key aims. Firstly to digitize 3D models of underwater artefacts for archaeological studies and preserving heritage with the use of affordable photogrammetric techniques. Secondly, It focuses on the use of video files as a data set and then explores the importance of adding EXIF data to frame captures from a video. This project uses and compares two free photogrammetric softwares which are Meshroom which is an open-source software and Autodesk Recap which provides free educational licence.

This underwater photogrammetry project uses images taken by the NIO, Goa team on an expedition to Dwarka in February of 2022. The entire data set included images and videos of sunken port, anchors and broken pillars.

This project does not include the data acquisition phase, it tries to create the best possible 3D models of underwater artefacts whose images and videos were taken before the commencement of this project. However intensive reading was done for data acquisition techniques in photogrammetry and suggestions have been made in the 7th section.

2. Literature Review

Over the past decade several studies have demonstrated the use of photogrammetry in archaeology. These studies have mainly used Agisoft Metashape (previously Agisoft Photoscan) and Autodesk Recap (Previously Autodesk 123DCatch). These studies extensively described the techniques and strategies used in the data acquisition phase which has been condensed and written as recommendations in the 7th section.

Much of the current literature on photogrammetry in archaeology uses images as the data set and follows a similar methodology in the data processing phase. In 2014, Bernardes et al. published a paper in which they demonstrated the use of image-based photogrammetry in archaeology. Their study involved the 3D reconstruction of the quadrangular pool of the Roman Villa in Milreu (Faro, Portugal) and The *Fonte do Ídolo* of *Bracara Augusta* (Braga, Portugal) using two photogrammetric softwares; VisualSfM and Autodesk 123DCatch. The comparative study between the two softwares showed that Autodesk 123DCatch was a more suitable choice. In 2018 Duong N. did a rigorous and detailed comparison between Agisoft Photoscan and Autodesk Recap Pro and his results showed that Autodesk Recap Pro provided a straightforward and streamlined user experience but gave very little control to the user to modify the 3D model and is more suitable for beginners. Agisoft Photoscan on the other hand gives users the control to modify the 3D model and is suitable for intermediate photogrammetrists.

Prior studies have not shown the use of video files as the data set for photogrammetry and the importance of EXIF data. In the underwater environment it can be quite challenging to shoot several quality images. Thus using videos as a data set can prove to be more convenient. Though the user manual of Agisoft Metashape states that the EXIF data, especially the focal length and the lens type is important for data processing and Meshroom software warns when images without these data are used, no study has yet demonstrated using frame captures from a video and adding the EXIF data to it.

3. Hardware Used

Minimum system requirements for AutoDesk Recap Pro is given below:

Operating System	Microsoft® Windows® 10 (64-bit)
CPU Type	2.0 gigahertz (GHz) or faster 64-bit (x64) processor
RAM Memory	8 GB
Display Resolution	1600 x 1050 or higher true colour
GPU	OpenGL 3.3 capable workstation-class graphics card with 1 GB or greater graphics memory
Browser	Google Chrome™ - latest version

Minimum system requirements for Meshroom is given below:

Operating System	Windows x64, Linux
CPU Type	Recent Intel or AMD CPUs
RAM Memory	8 GB
GPU	NVIDIA CUDA-enabled GPU (compute capability >= 2.0)

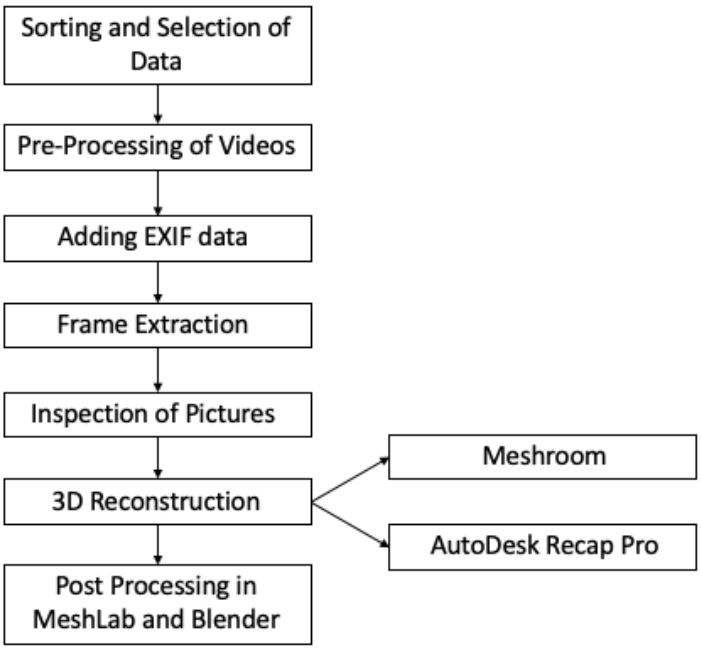
The processing time when using AutoDesk Recap Pro will largely depend on the internet speed. However the processing time while using Meshroom will depend on the hardware specifications. It is essential to use NVIDIA CUDA-enabled GPU with Meshroom for creating depth maps.

Razer Blade 14 with the following specifications was used for the final data processing:

Operating System	Windows 11 Home
CPU	AMD Ryzen 9 5900HX
RAM	16 GB
System Type	64-bit operating system
GPU	NVIDIA GeForce RTX 3070

4. Data Processing

The data processing phase involved the following manual and automated steps:



A. Step 1: Manual sorting and selection of videos

The large set of data (around 102 GB) was manually inspected to find reliable images and videos which can be used for 3D reconstruction. After sorting the images and videos according to the respective artefacts it was found that there were not enough reliable images for 3D reconstruction, this meant that frame captures from videos had to be used. Six artefacts were identified and they were named “artefact_x” (x being the

number of the artefact between 1 and 6).Images and videos were sorted into six different folders. After the manual sorting artefacts 2 and 4 did not have enough images or videos that could yield good results. Videos of artefact 1, 3, 5 and 6 were chosen as shown in fig.1

Videos selected had the following features:

- The object covered the maximum area in the frame.
- The video was stable
- The video did not have varying zoom.
- The video must not be an extremely closeup video.
- The video covered most if not all the sides of the object.
- If 2 different videos are being used then it was make sure the frames from the two videos have at least 50% overlap.

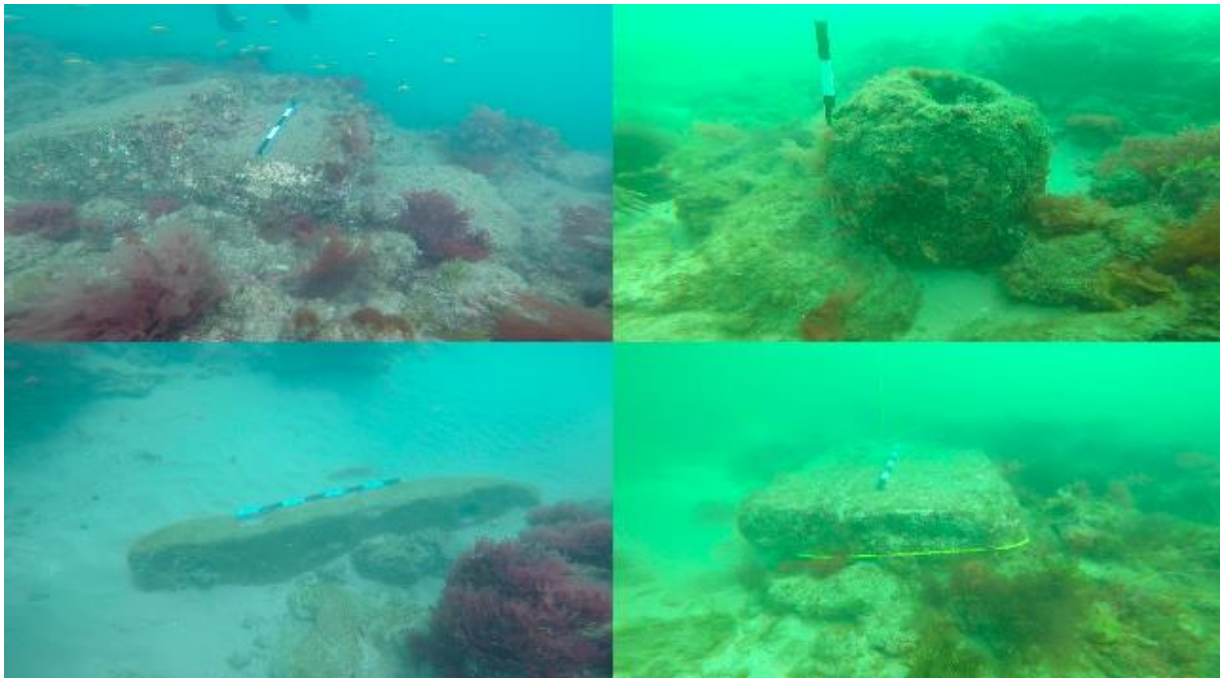


Fig. 1 Selected artefacts (Top left – artefact 1, top right – artefact 3, bottom left – artefact 5, bottom right – artefact 6)

B. Step 2: Pre-processing of videos

Due to loss of long wavelength colours like red, blue/green dominates underwater images and videos hence colour correction is important to restore the natural colour of the artefact. For this purpose Adobe premiere pro was used. Figure 2 shows a comparison between the original and colour corrected videos.

- Red and purple colour was added to reduce the blue/green hue.
- Contract can be adjusted to get sharper images but too much contrast can overexpose some surfaces and create noise in the 3D reconstruction process.
- Parts of the videos that feature large moving objects like humans can be trimmed.



Fig.2 Original videos (Left) Color corrected videos (Right)

C. Step 3: EXTRACTING FRAMES FROM VIDEOS USING VLC

VLC media player was used to capture the frame from the videos by following the given steps:

- Open VLC media player
- VLC > Preferences > Show all > Video > Filters > Check *scene video filter*
- Drop down filters > Scene filter > Set the desired values as exemplified in fig. 3.
- Save and relaunch VLC media player.
- Play the desired video and the frames will be saved to the mentioned directory
- After the frames have been captured uncheck the *scene video filter*.

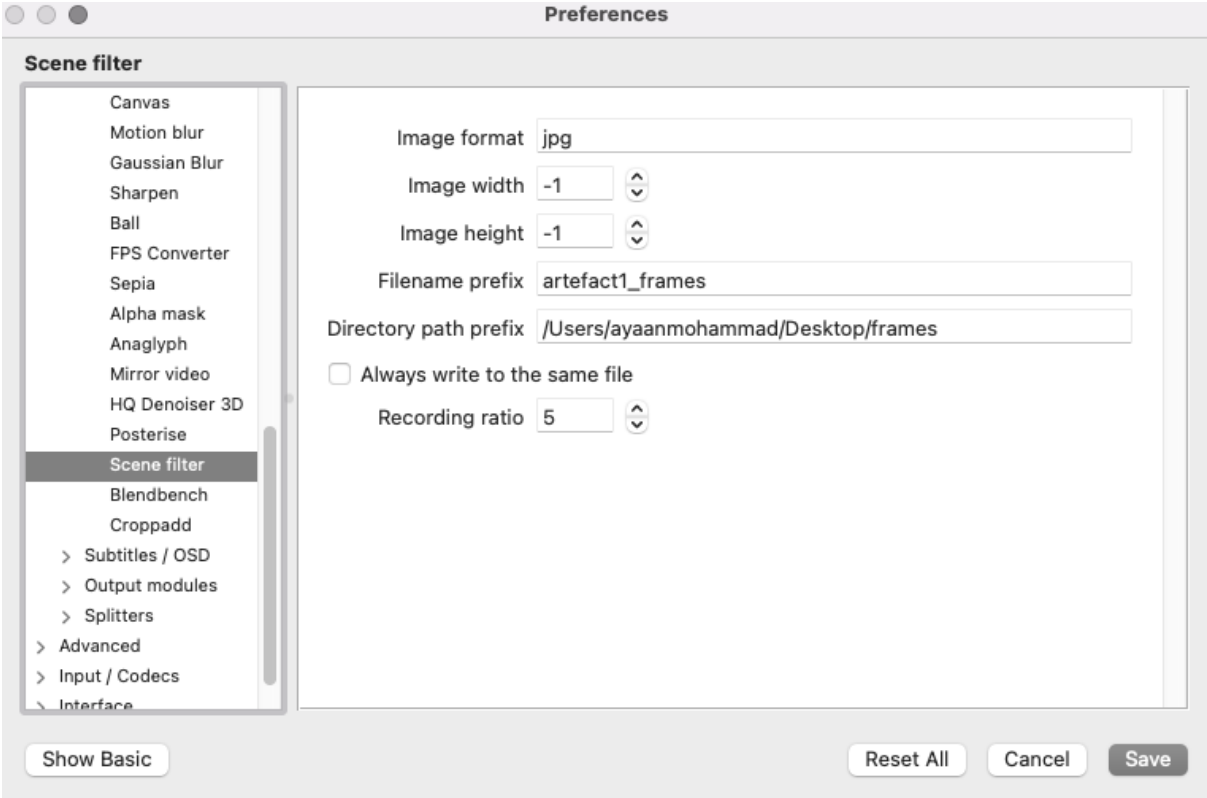


Fig. 3 Preferences for scene capture on VLC

D. Step 4: ADDING EXIF DATA TO CAPTURED FRAMES

EXIF(Exchangeable Image File Format) stores data about the photograph such as focal length, location, camera make, lens type etc.

After capturing the frames, the set of captured frames were duplicated and the known focal length was added to one set of images while no EXIF data was added to the other set. This was done to compare the results between the set of images with the EXIF data and the set of images without EXIF data.

Steps to add EXIF data:

- Download exiftool from https://exiftool.org/
- Install exiftool
- Paste a shell script to the directory with the captured frames containing the code:

```
#!/bin/bash
for i in *.jpg
do
  exiftool -Focallength=3 ${i}
done
```

- Open terminal > make the directory with the captured frames and .sh file(a.sh) your current directory
- Write ./a.sh > enter
- If the ISO, Aperture And F-stop is known then they can be added as well.

E. Step 5: Manual Inspection of the pictures

- Look for images that are blurry and too much reflection or overexposed and delete them.
- Agisoft Metashape which is a paid photogrammetric software gives an option to automate this step.

F. Step 6: 3D reconstruction in Meshroom/Autodesk Recap Pro

Workflow of Meshroom:

- Open meshroom > Import the images
- Meshroom requires CUDA and a NVIDIA graphic card for depth maps. Without CUDA it can create a draft mesh. https://github.com/alicevision/meshroom/wiki/Draft-Meshing
- If you do not have a NVIDIA graphic card then delete the PrepareDenseScene, DepthMap and DepthMapFilter nodes. And Join the StructurefromMotion to Meshing node. Figure 4 shows the node settings if NVIDIA card is not being used.
- Save the Project > Click start button
- This will start the Photo aligning, Point cloud generation, Meshing and Texturing process.
- After the process is completed go to the project folder > Texturing Folder and open the .obj file in MeshLab.

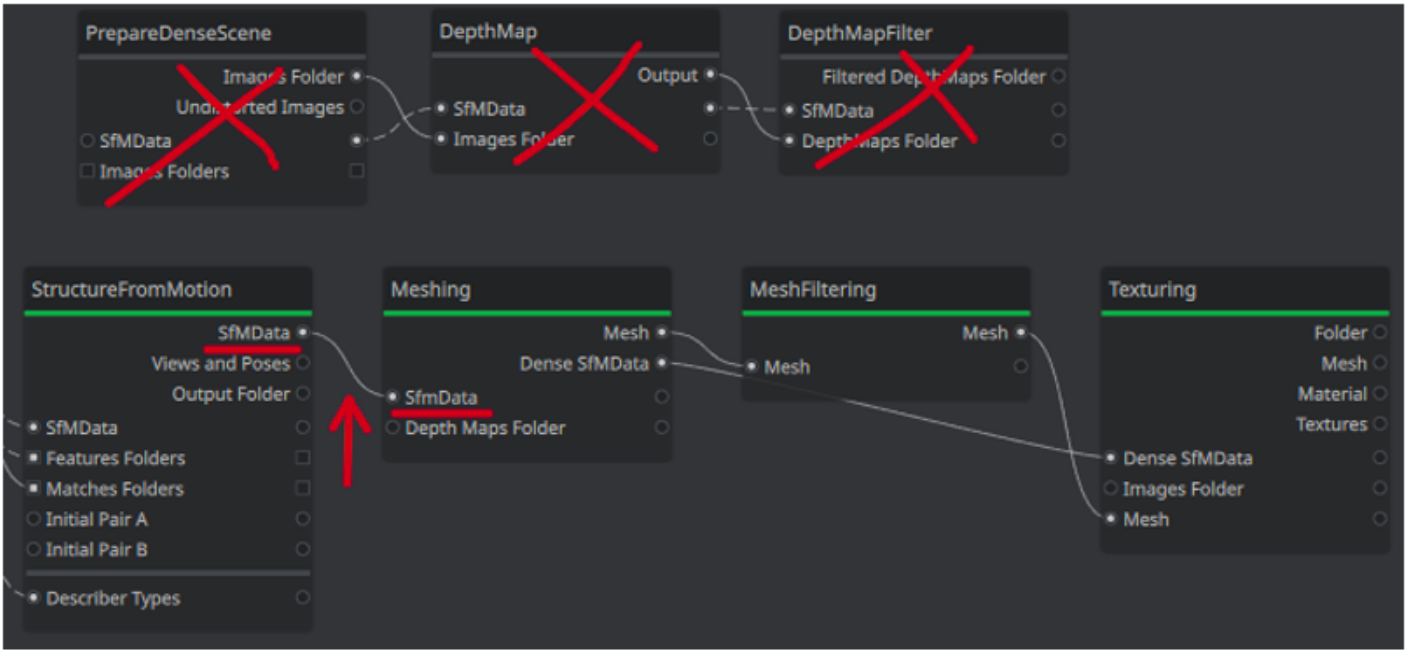


Fig. 4 Node setup for processing without an Nvidia card

Workflow of Autodesk Recap Pro:

- Open Autodesk Recap
- New Project > Photo to 3D
- Recap Photo > Object
- Click anywhere to browse files > select the required files> Create
- Give project name> connect to a directory in Autodesk Drive > Start
- This starts the 3D reconstruction process, the sparse point cloud, dense point cloud, meshing and texturing process are fully automated in Autodesk Recap.
- Once the 3D reconstruction is completed, the 3D model can be edited in Autodesk Photo.
- Export as model(.obj) > Same the files in a single directory.

G. Step 7: Post-processing in MeshLab

MeshLab is an open-source software for processing and editing 3D triangular meshes. It provides tools for editing, cleaning, healing, inspecting, rendering, texturing and converting meshes.

The following steps can be followed to work with a mesh in MeshLab:

- Import mesh > select the .obj file (make sure that the directory has the .obj file, the texture .png file and the .obj.mtl file)
- Remove the unnecessary vertices and faces.
- Simplify, repair or smoothen the mesh using the various filter from the filters dropdown menu.
- Export mesh > uncheck the *normal* box > export.

H. Step 8: Post-processing in Blender

Blender is an open-source 3D computer graphics software that can be used for many 3D applications.

- Import > .obj file
- Measure the model and scale as required
- Improve the model with sculpting and texturing tools.
- An animation of the final model can be made using blender.

5. Result and Discussion

Multiple 3D models were made of the 4 selected artefacts using both Meshroom and Autodesk Recap Pro. Two different sets of the same images were, one with focal length added to the EXIF data and one without that. The results showed that adding focal length to the images significantly improved the image matching process and reduced floating vertices.

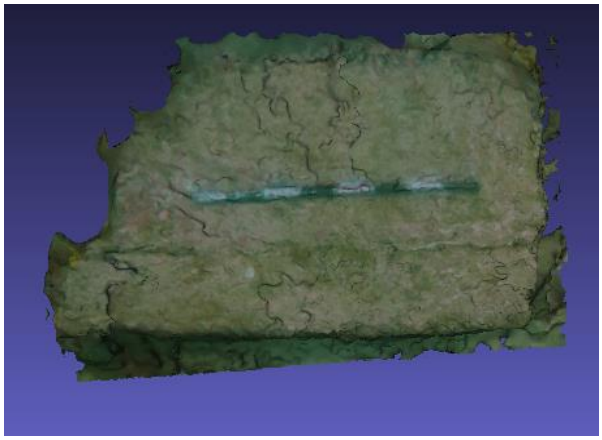


Fig. 5 Processed using images with focal defined as 3mm (Focal length of a GoPro camera)

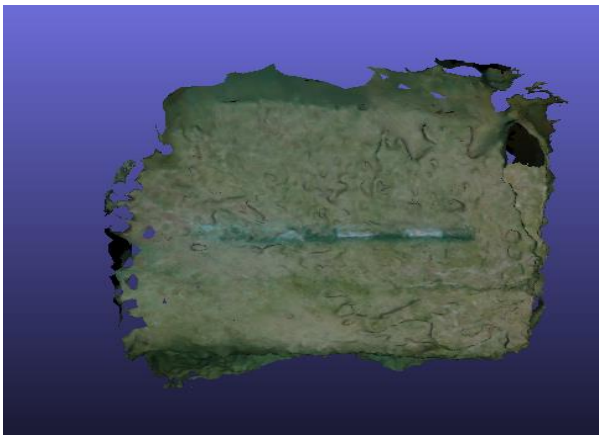
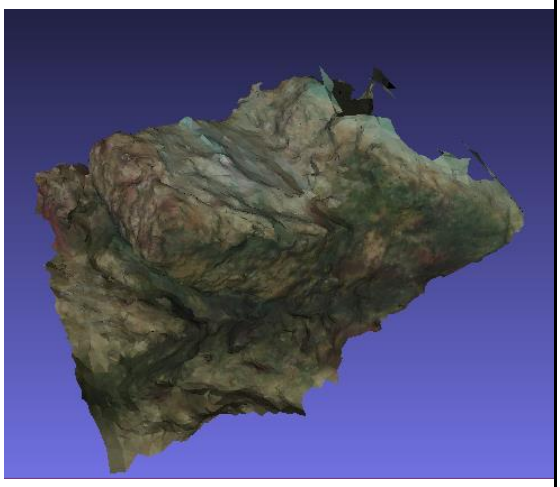
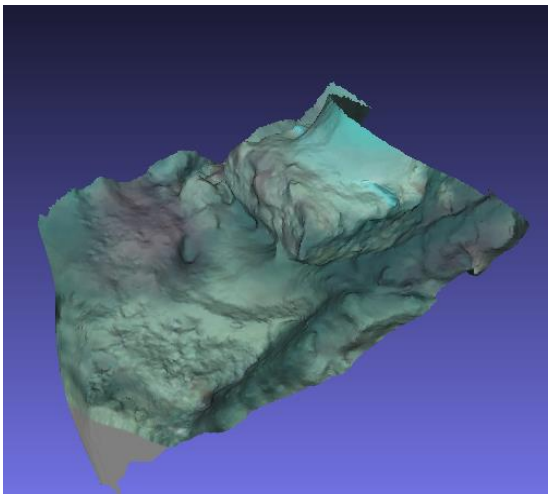
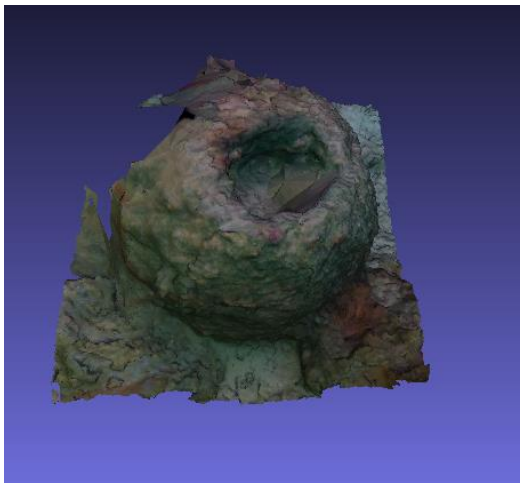
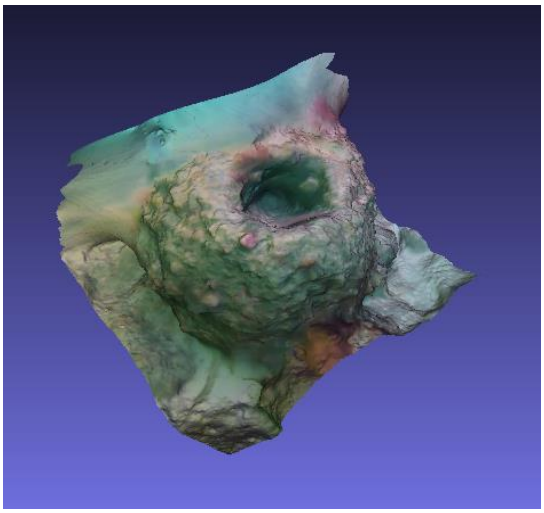
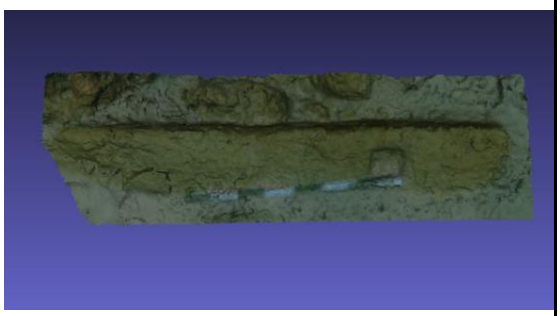
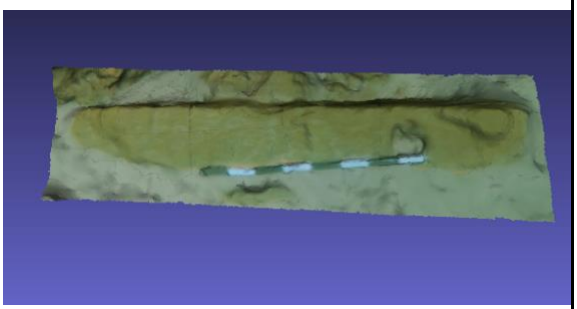
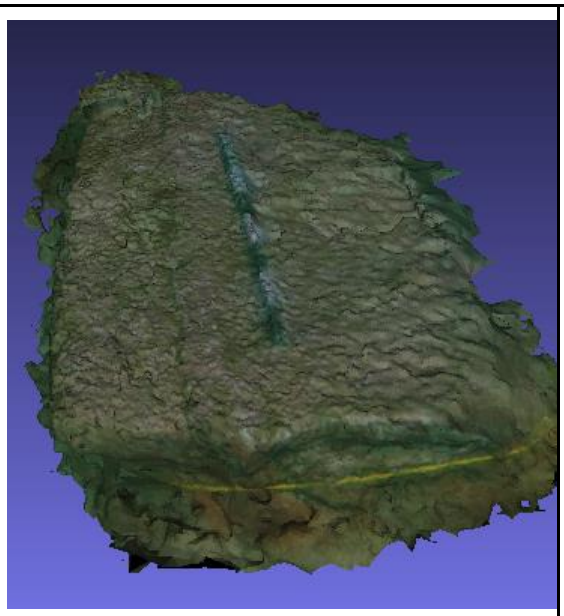
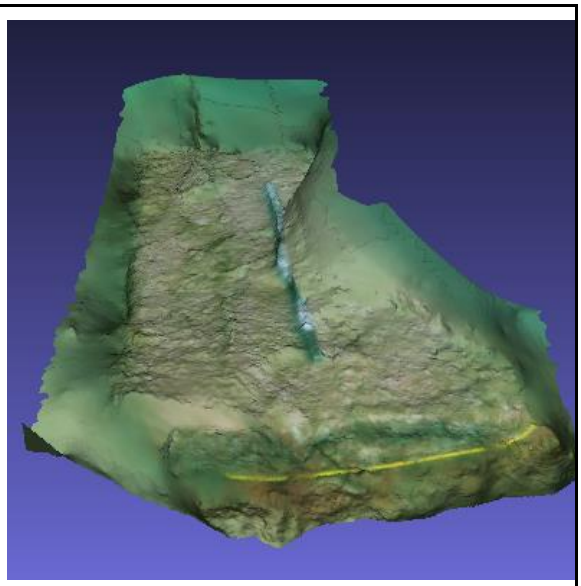


Fig. 6 Processed using images that lacked

The following table shows a comparison of 3D models produced using Meshroom versus Autodesk Recap Pro

	Meshroom	Autodesk Recap Pro
Artefact 1		
Artefact 3		
Artefact 5		

Artefact 6		

Comparing the models from the two photogrammetric softwares, we can see that Meshroom has better edge detection and does not create vertices due to turbidity in water. However models created in Meshroom have many bumps and floating vertices which needs more cleaning and post-processing work. Models created on Autodesk Recap Pro have a smoother and a more realistic surface with nearly zero floating vertices. In terms of texture Autodesk Recap Pro creates much better texture as compared to Meshroom.

6. Conclusion

After studying about photogrammetric processes and evaluating the results we can draw the following conclusions:

- Still images are a far better data set for a photogrammetric project. However if frames from a video is being used adding EXIF data to the set of captures frames is essential and significantly affects your final results.
- Meshroom creates more defined models but requires CUDA and Nvidia Graphics card for creating Depth maps and takes a significantly longer time for processing as compared to Autodesk Recap Pro. The texture created by Meshroom can be improved by some colour correction using a photo editing software like Adobe Lightroom.
- Meshroom can significantly increase your post-processing time but it gives more control on the process to the user.
- Autodesk Recap Pro creates models which requires nearly zero post-processing. It does not require sophisticated hardware as the entire Photogrammetric process is cloud based. Your processing time will depend on your internet speed. However Autodesk Recap Pro does not give much control on the photogrammetric process to the user.

7. Future Work

In underwater archaeological sites like Dwarka where the artefacts are spread over a large area, the project can be divided into two phases. First, the archaeological site must be photographed according to photogrammetric needs followed by 3D reconstruction of those artefacts. Geo-referenced photogrammetry can be very useful in underwater archaeological sites as data acquisition of underwater sites can take multiple days and the environmental conditions may vary significantly.

A better dataset can have the most significant improvement in the quality of the models. The following guidelines can be followed for better data acquisition:

- An overlap of 60% is recommended.
- Wide angle camera lenses must be avoided to reduce distortion of images.
- It is recommended to use lens of fixed focal length ranging between 20-80 mm.
- If using a zooming lens the focal length should be fixed to the maximum or minimum focal length.
- The target object must cover the maximum area in the frame.
- No source of light must be present in the frame.
- Pictures must be taken on overcast days to reduce glare are reflection.
- If you are planning to carry out any measurements based on the reconstructed model, do not forget to locate at least two markers with a known distance between them on the object. Alternatively, you could place a ruler within the shooting area.
- Do not crop or geometrically transform the image.
- Snake or spiral pattern is recommended for underwater data acquisition
- Coloured control points with irregular pattern must be placed over the target area.

References

1. Zhukovsky, M.O., 2013. PHOTOGRAMMETRIC TECHNIQUES FOR 3–D UNDERWATER RECORD OF THE ANTIQUE TIME SHIP FROM PHANAGORIA.
2. Bernardes, P., Magalhães, F., Ribeiro, J., Madeira, J. and Martins, M., 2014. Image-based 3D modelling in archaeology: application and evaluation.
3. Diamanti, E. and Vlachaki, F., 2015. 3D Recording of underwater antiquities in the South Euboean Gulf. *The International Archives of Photogrammetry, Remote Sensing and Spatial Information Sciences*, *40*(5), p.93.
4. Balletti, C., Beltrame, C., Costa, E., Guerra, F. and Vernier, P., 2015. Underwater Photogrammetry and 3D Reconstruction of Marble Cargos Shipwreck. *International Archives of the Photogrammetry, Remote Sensing & Spatial Information Sciences*.
5. Duong, N., 2018. A comparative case study between Agisoft Photoscan and Autodesk Recap Photo-To-3D web services.
6. Abdelaziz, M. and Elsayed, M., 2019. UNDERWATER PHOTOGRAMMETRY DIGITAL SURFACE MODEL (DSM) OF THE SUBMERGED SITE OF THE ANCIENT LIGHTHOUSE NEAR QAITBAY FORT IN ALEXANDRIA, EGYPT. *International Archives of the Photogrammetry, Remote Sensing & Spatial Information Sciences*.
7. (Aligning The Images – Why Things Go Wrong, 2019)
8. Jones, C.A. and Church, E., 2020. Photogrammetry is for everyone: Structure-from-motion software user experiences in archaeology. *Journal of Archaeological Science: Reports*, *30*, p.102261.
9. Lee, Maping Life on Seafloor, 2020
10. Griwodz, C., Gasparini, S., Calvet, L., Gurdjos, P., Castan, F., Maujean, B., De Lillo, G. and Lanthony, Y., 2021, June. AliceVision Meshroom: An open-source 3D reconstruction pipeline. In *Proceedings of the 12th ACM Multimedia Systems Conference* (pp. 241-247).
11. (Agisoft Metashape User Manual: Standard Edition, Version 1.8, 2022)
12. (Photogrammetry: Step-by-Step Guide and Software Comparison, n.d.)